

第7章 频率特性和谐振现象



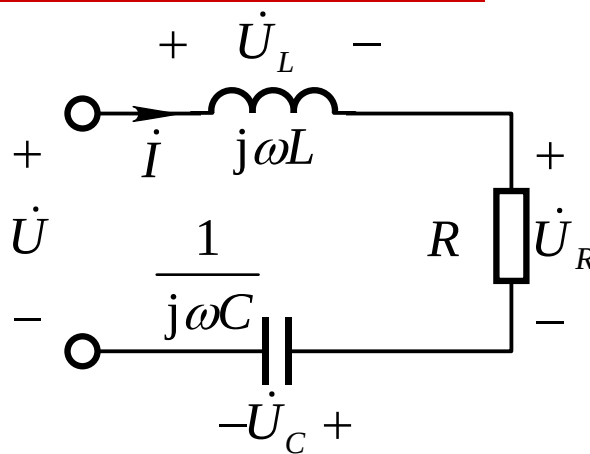
7.3 RLC 串联电路的频率特性

基本要求：了解 RLC 串联电路的网络函数及其频率特性。

主要内容

- 一、以电阻电压为响应的网络函数
- 二、以电容电压为响应的网络函数
- 三、以电感电压为响应的网络函数

7.3 RLC 串联电路的频率特性



$$H_R(j\omega) = \frac{\dot{U}_R}{\dot{U}} = \frac{R}{R + j[\omega L - 1/(\omega C)]}$$

$$H_C(j\omega) = \frac{\dot{U}_C}{\dot{U}} = \frac{1/(j\omega C)}{R + j\omega L + 1/(j\omega C)}$$

$$H_L(j\omega) = \frac{\dot{U}_L}{\dot{U}} = \frac{j\omega L}{R + j\omega L + 1/(j\omega C)}$$

一、以电阻电压为响应的网络函数

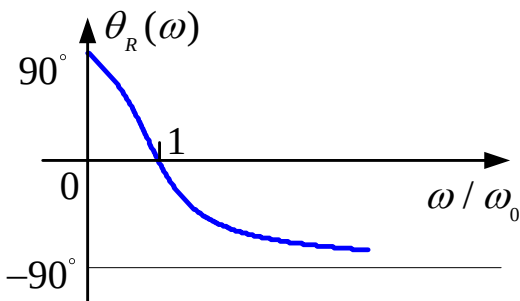
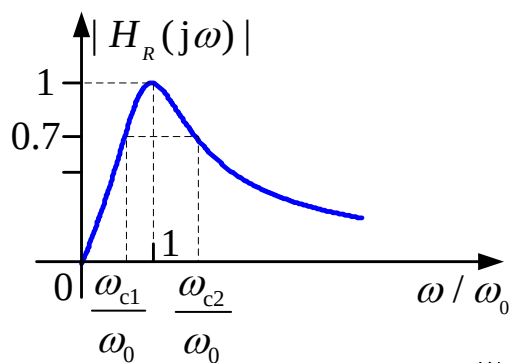
$$H_R(j\omega) = \frac{\dot{U}_R}{\dot{U}} = \frac{R}{R + j[\omega L - 1/(\omega C)]}$$

$$|H_R(j\omega)| = \frac{1}{\sqrt{1 + \frac{1}{R^2} \left(\omega L - \frac{1}{\omega C} \right)^2}} = \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$

$$\theta_R(\omega) = -\arctan Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

ω / ω_0	$ H_R(j\omega) $	$\theta_R(\omega)$
0	0	90°
1	1	0°
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
∞	0	-90°

一、以电阻电压为响应的网络函数



*RLC*带通电路的频率特性

$$|H_R(j\omega)| = \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$

$$\theta_R(\omega) = -\arctan Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

1. $|H_R(j\omega)|$ 具有带通特性

$$\frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega_c}{\omega_0} - \frac{\omega_0}{\omega_c} \right)^2}} = \frac{1}{\sqrt{2}}$$

截止
频率

$$\omega_{c1} = \omega_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$

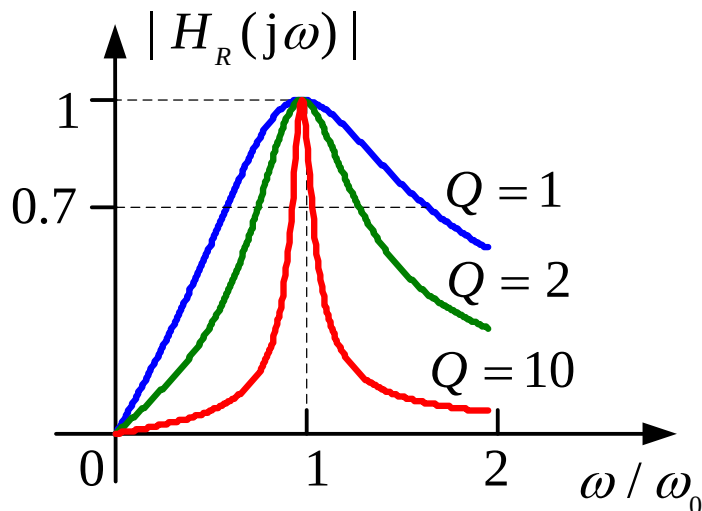
$$\omega_{c2} = \omega_0 \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$

通带宽度

$$\Delta\omega = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q}$$

一、以电阻电压为响应的网络函数

2. 频率特性与品质因数的关系



$$\Delta\omega = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q}$$

(1) Q 值越大，截止频率处的曲线越陡，频率**选择性越好，带宽越窄**。

(2) Q 值越小，带宽越宽，选择性能越差。

RLC 串联电路，频率选择性与带宽存在矛盾。

二、以电容电压为响应的网络函数

$$H_C(j\omega) = \frac{\dot{U}_C}{\dot{U}} = \frac{1/(j\omega C)}{R + j\omega L + 1/(j\omega C)}$$

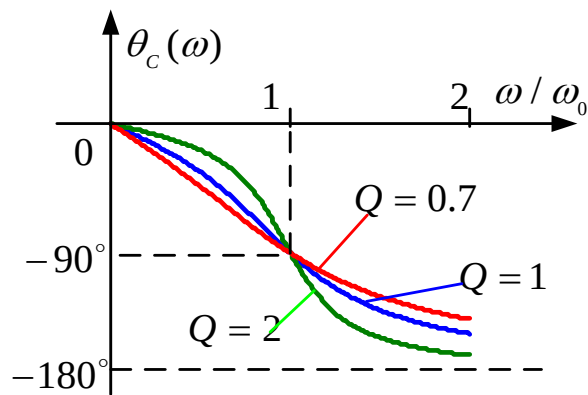
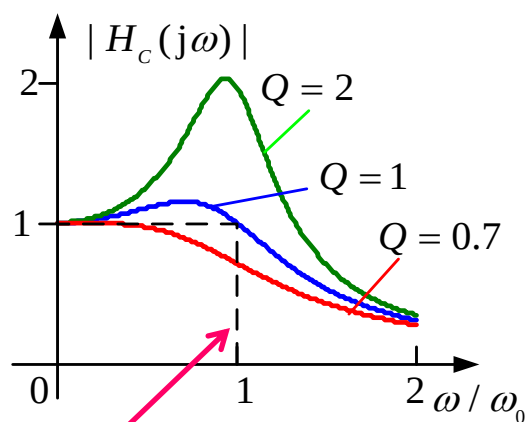
$$= \frac{1}{(1 - \omega^2 LC) + j\omega CR} = \frac{1}{[1 - \left(\frac{\omega}{\omega_0}\right)^2] + j\frac{1}{Q}\left(\frac{\omega}{\omega_0}\right)}$$

$$|H_C(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2}\left(\frac{\omega}{\omega_0}\right)^2}}$$

$$\theta_C(\omega) = -\arctan \frac{1}{Q\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)}$$

ω / ω_0	$ H_C(j\omega) $	$\theta_C(\omega)$
0	1	0°
1	Q	-90°
⋮	⋮	⋮
⋮	⋮	⋮
∞	0	-180°

二、以电容电压为响应的网络函数



RLC低通电路的滤波特性

理想情况

$|H_C(j\omega)|$ 具有低通特性

$$|H_C(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega}{\omega_0}\right)^2}}$$
$$\theta_C(\omega) = -\arctan \frac{1}{Q \left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)}$$

三、以电感电压为响应的网络函数

$$H_L(j\omega) = \frac{\dot{U}_L}{\dot{U}} = \frac{j\omega L}{R + j\omega L + 1/(j\omega C)}$$

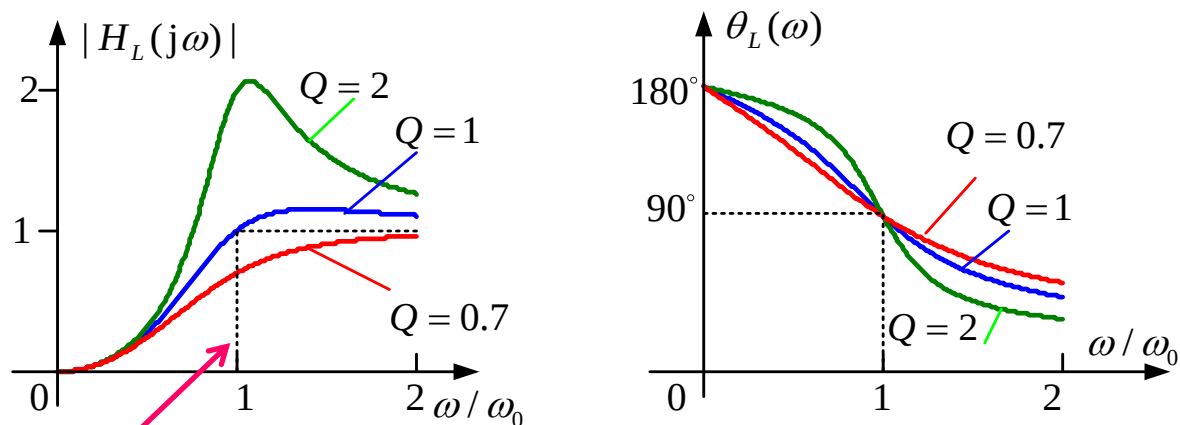
$$= \frac{1}{[1 - 1/(\omega^2 LC)] - jR/(\omega L)} = \frac{1}{\left[1 - \left(\frac{\omega_0}{\omega}\right)^2\right] - j\frac{1}{Q}\left(\frac{\omega_0}{\omega}\right)}$$

$$|H_L(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega_0}{\omega}\right)^2\right]^2 + \frac{1}{Q^2}\left(\frac{\omega_0}{\omega}\right)^2}}$$

$$\theta_L(\omega) = -\arctan \frac{-1}{Q\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

ω / ω_0	$ H_L(j\omega) $	$\theta_L(\omega)$
0	0	180°
1	Q	90°
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
∞	1	0°

三、以电感电压为响应的网络函数



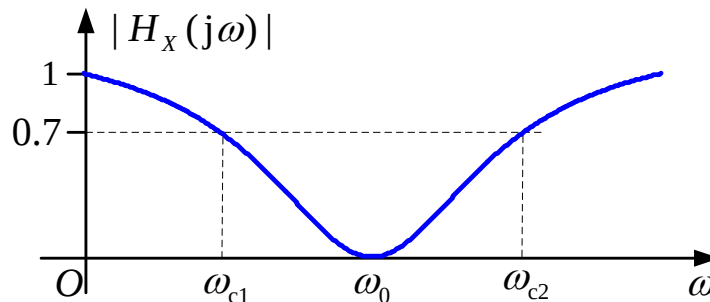
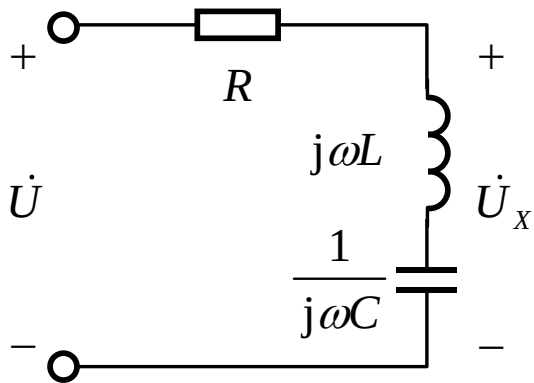
RLC高通电路频率特性

理想情况

$|H_L(j\omega)|$ 具有高通特性

$$|H_L(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega_0}{\omega}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega_0}{\omega}\right)^2}}$$
$$\theta_L(\omega) = -\arctan \frac{-1}{Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)}$$

思考：RLC串联电路能否实现带阻特性？



$$H_X(j\omega) = \frac{\dot{U}_X}{\dot{U}} = \frac{j[\omega L - 1/(\omega C)]}{R + j[\omega L - 1/(\omega C)]}$$

$$\omega_{c1} = \omega_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$

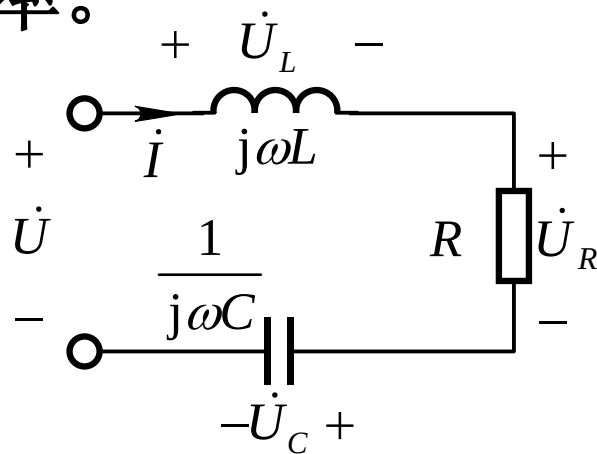
$$\omega_{c2} = \omega_0 \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$

【例题7.3】

设计 RLC 带通电路，已知总电阻为 $R = 20\Omega$ ，要求谐振频率为 $f_0 = 10^4 \text{ Hz}$ ，带宽为 $\Delta f = 10^3 \text{ Hz}$ ，求电感 L 和电容 C 的值以及低频截止频率和高频截止频率。

【解】

$$\Delta\omega = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q} \longrightarrow Q = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f} = 10$$

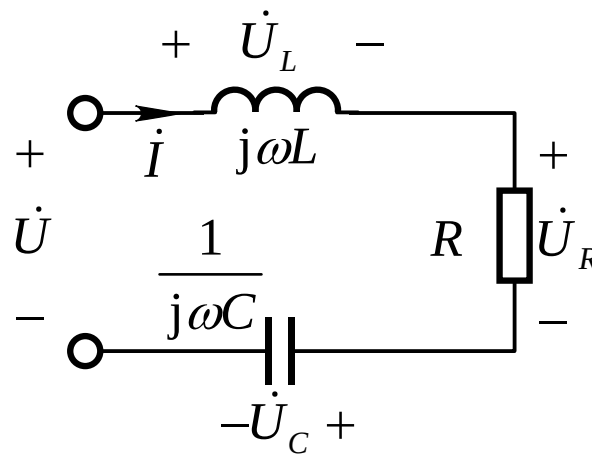


$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} \longrightarrow L = \frac{QR}{\omega_0} = \frac{10 \times 20\Omega}{(2\pi \times 10^4) \text{ s}^{-1}} \approx 3.18 \text{ mH}$$

$$C = \frac{1}{QR\omega_0} = \frac{1}{10 \times 20\Omega \times (2\pi \times 10^4) \text{ s}^{-1}} \approx 0.0796 \mu\text{F}$$

【例题7.3】

$$\omega_{c1} = \omega_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$
$$\omega_{c2} = \omega_0 \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$



$$f_{c1} = f_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$
$$\approx f_0 \left(-\frac{1}{2Q} + 1 \right) = 9500\text{Hz}$$

$$f_{c2} = f_0 \left(+\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$
$$\approx f_0 \left(+\frac{1}{2Q} + 1 \right) = 10500\text{Hz}$$

【补充7.10】

RLC 串联电路的谐振频率为875Hz，通带宽度为250Hz，已知 $L = 0.32\text{H}$ ：

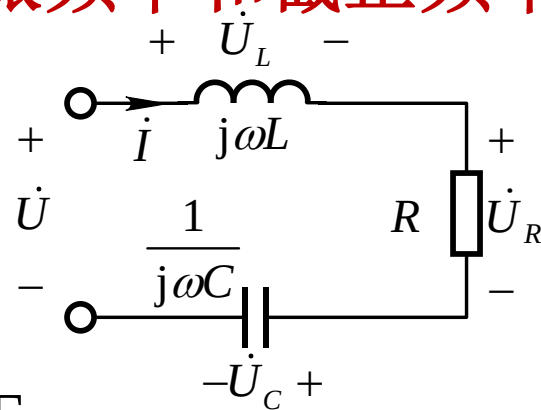
- (1). 求 R 、 C 、品质因数 Q 及截止频率；
- (2). 设输入电压有效值为23.2V，求谐振频率和截止频率时电路的平均功率；
- (3). 求谐振时电感电压和电容电压。

【解】 (1). 求 R 、 C 、 Q 、截止频率

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi \times 875)^2 \times 0.32} = 1.034 \times 10^{-7} \text{ F}$$

$$\Delta f = \frac{f_0}{Q} \Rightarrow Q = \frac{f_0}{\Delta f} = 875 / 250 = 3.5$$

$$Q = \frac{\omega_0 L}{R} \Rightarrow R = \frac{\omega_0 L}{Q} = 2\pi \times 875 \times 0.32 / 3.5 = 502.65 \Omega$$

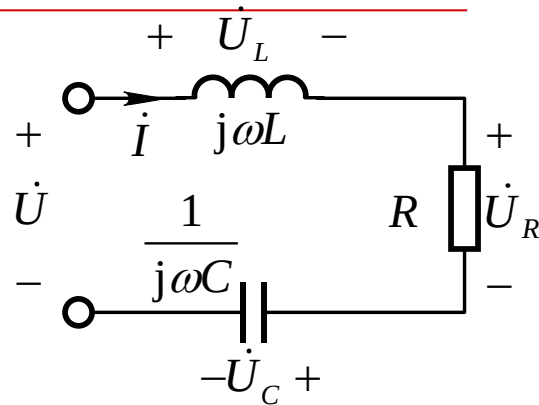


【补充7.10】

截止频率:

$$f_{c1} = \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1}\right) \times f_0 \approx 759\text{Hz}$$

$$f_{c2} = \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1}\right) \times f_0 \approx 1009\text{Hz}$$



(2). 设输入电压有效值为**23.2V**，求**谐振频率**和**截止频率**时电路的**平均功率**；

谐振频率时的**平均功率**: $P_0 = I_0^2 R$
 $= (23.2 / 502.65)^2 \times 502.65 = 1.071\text{W}$

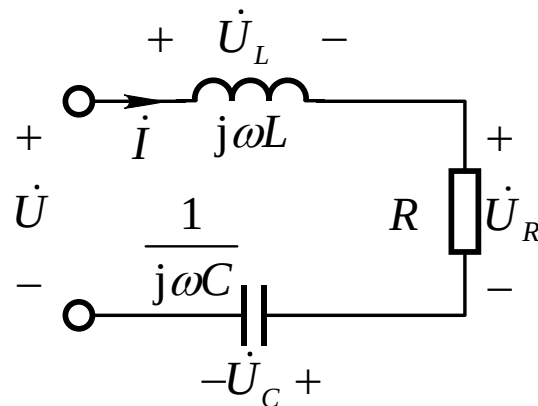
在截止频率处，电流下降至谐振电流 I_0 的 $1/\sqrt{2}$ ，故功率减小到一半。

截止频率时的**平均功率**: $P = P_0 / 2 = 0.535\text{W}$

【补充7.10】

(3).谐振时电感电压和电容电压:

$$U_L = U_C = QU = 3.5 \times 23.2 = 81.2\text{V}$$



【补充7.11】

电路中，已知： $L_1 = 0.01\text{H}$ $R_1 = 5\Omega$ $L_2 = 0.02\text{H}$ $R_2 = 10\Omega$

$$M = 0.01\text{H} \quad C = 20\mu\text{F}$$

- 1、求两线圈顺接、反接时的谐振角频率和带宽。
- 2、两种情况下外加电压均为6V，试求两线圈上的电压 U_1 和 U_2 。

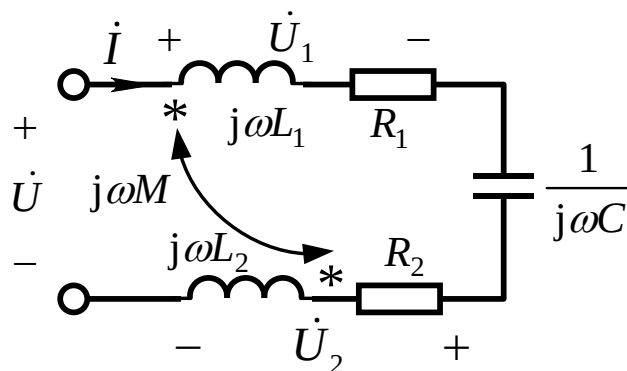
【解】 当两线圈顺接时

$$L = L_1 + L_2 + 2M = 0.05\text{H}$$

谐振角频率

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.05 \times 20 \times 10^{-6}}} = 10^3 \text{ rad/s}$$

$$\text{带宽 } \Delta\omega = \frac{\omega_0}{Q} = \frac{R_1 + R_2}{L} = \frac{15}{0.05} = 300 \text{ rad/s}$$

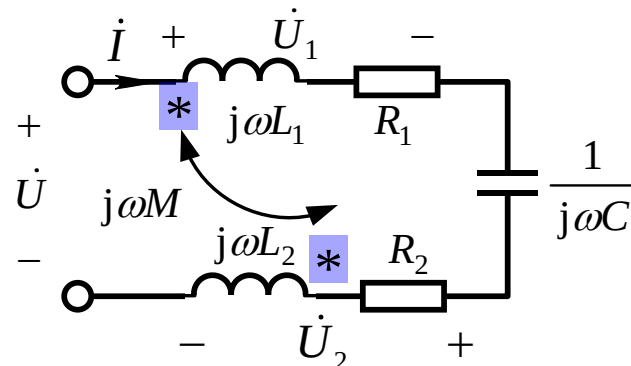


【补充7.11】

取 $\dot{U} = 6\angle 0^\circ \text{V}$

谐振时的电流:

$$\dot{I} = \frac{\dot{U}}{R_1 + R_2} = \frac{6\angle 0^\circ}{5 + 10} \text{A} = 0.4\angle 0^\circ \text{A}$$



$$\dot{U}_1 = (R_1 + j\omega_1 L_1)\dot{I} + j\omega_1 M\dot{I} = [(5 + j10) \times 0.4 + j10 \times 0.4] \text{V} = (2 + j8) \text{V}$$

$$\dot{U}_2 = (R_2 + j\omega_1 L_2)\dot{I} + j\omega_1 M\dot{I} = [(10 + j20) \times 0.4 + j10 \times 0.4] \text{V} = (4 + j12) \text{V}$$

$$U_1 = \sqrt{2^2 + 8^2} = 8.24 \text{V} \quad U_2 = \sqrt{4^2 + 12^2} = 12.65 \text{V}$$

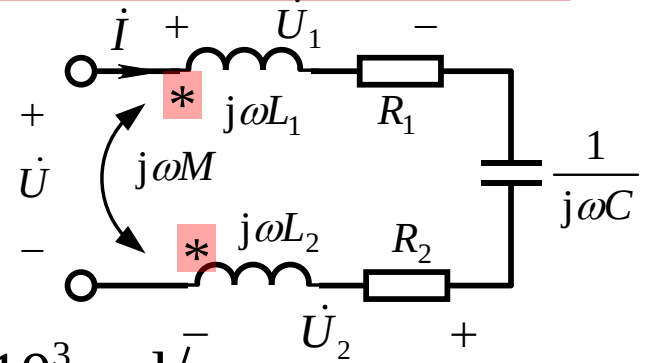
【补充7.11】

当两线圈反接时:

$$L = L_1 + L_2 - 2M = 0.01\text{H}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.01 \times 20 \times 10^{-6}}} = 2.236 \times 10^3 \text{ rad/s}$$

$$\Delta\omega = \frac{\omega_0}{Q} = \frac{R_1 + R_2}{L} = \frac{15}{0.01} = 1500 \text{ rad/s}$$



取 $\dot{U} = 6\angle 0^\circ \text{V}$ 时

谐振电流:
$$\dot{I} = \frac{\dot{U}}{R_1 + R_2} = \frac{6\angle 0^\circ}{5 + 10} \text{ A} = 0.4\angle 0^\circ \text{ A}$$

$$\dot{U}_1 = (R_1 + j\omega_2 L_1)\dot{I} - j\omega_2 M\dot{I} = 5\Omega \times 0.4\text{A} = 2\text{V}$$

$$\dot{U}_2 = (R_2 + j\omega_2 L_2)\dot{I} - j\omega_2 M\dot{I} = (10 + j22.36)\Omega \times 0.4\text{A} = (4 + j8.95)\text{V}$$

$$U_1 = 2\text{V} \quad U_2 = \sqrt{4^2 + 8.95^2} = 9.8\text{V}$$